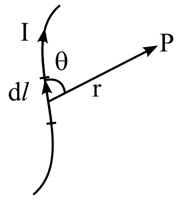
Moving Charges and Magnetism

SHORT NOTES

MAGNETIC FIELD PRODUCED BY A CURRENT (BIOT-SAVARTS LAW)

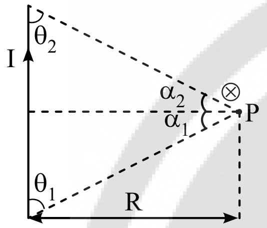
$$dB = \frac{\mu_0\mu_r}{4\pi}\frac{I\,dl\sin\theta}{r^2} \Rightarrow \vec{dB} = \frac{\mu_o\mu_r}{4\pi}\frac{I\left(d\vec{l}\times\vec{r}\right)}{r^3}$$

here the quantity *Idl* is called as current element.



Magnetic Induction due to a Straight Current Conductor

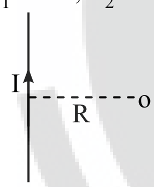
Magnetic induction due to a current carrying straight wire



$$B = \frac{\mu_0 I}{4\pi R}(\cos\theta_1 + \cos\theta_2) = \frac{\mu_0 I}{4\pi R}(\sin\alpha_1 + \sin\alpha_2)$$

Magnetic induction due to a infinitely long wire $B = \frac{\mu_0 I}{2\pi R}\otimes$

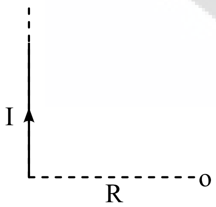
$\alpha_1 = 90^o$; $\alpha_2 = 90^o$



Magnetic induction due to semi infinite straight conductor

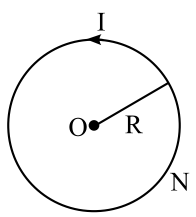
$B = \frac{\mu_0 I}{4\pi R}\otimes$

$\alpha_1 = 0^o$; $\alpha_2 = 90^o$



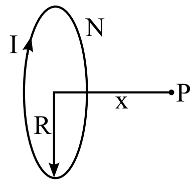
- Magnetic field due to a flat circular coil carrying a current:

(*i*) At its centre $B = \frac{\mu_0 NI}{2R}\odot$



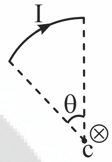
(*ii*) On the axis $B = \frac{\mu_0 NIR^2}{2\left(x^2+R^2\right)^{3/2}}$

It is maximum at the centre $B_C = \frac{\mu_0 NI}{2R}$



(*iii*) Magnetic field due to flat circular *ARC*:

$B = \frac{\mu_0 I\theta}{4\pi R}$



(*iv*) Magnetic field due to infinite long solid cylindrical conductor of radius *R*

- For $r \geq R$: $B = \frac{\mu_0 I}{2\pi r}$
- For $r < R$: $B = \frac{\mu_0 Ir}{2\pi R^2}$

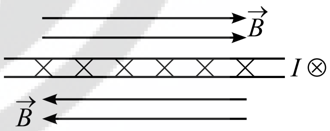
Magnetic Induction due to Solenoid

$B = \mu_0 nI$, direction along axis.

$B = \frac{\mu_0 nI}{2}$, at the end point.

Magnetic Induction due to Current Carrying Sheet

$B = \frac{1}{2}\mu_0 J$, where J = Linear current density (A/m)



Amperes Circuital Law

$\oint \vec{B}.\vec{d}l = \mu\Sigma I$ where ΣI = algebraic sum of all the current.

Motion of a Charge in Uniform Magnetic Field

(*a*) When $\vec{v} \parallel \vec{B}$; Motion will be in a straight line and $\vec{F} = 0$

(*b*) When $\vec{v} \perp \vec{B}$; Motion will be in circular path with radius $R = \frac{mv}{qB}$ and angular velocity $\omega = \frac{qB}{m}$ and $F = qvB$.

(*c*) When $\vec{v}$ makes θ with $\vec{B}$: Motion will be helical with radius $R = \frac{mv\sin\theta}{qB}$ and pitch $P = \frac{2\pi mv\cos\theta}{qB}$ and $F = qvB\sin\theta$.

Lorentz Force

$$\vec{F} = q(\vec{v}\times\vec{B}) + q\vec{E}$$

Motion of Charge in Combined Electric Field & Magnetic Field

- When $\vec{v}\parallel\vec{B}$ & $\vec{v}\parallel\vec{E}$, Motion will be uniformly accelerated in line as $F_{magnetic} = 0$ and $F_{electrostatic} = qE$
- When $\vec{v}\parallel\vec{B}$ & $\vec{v}\perp\vec{E}$, motion will be uniformly accelerated in a parabolic path
- When $\vec{v}\perp\vec{B}$ & $\vec{v}\perp\vec{E}$, the particle will move undeflected & undeivated with same uniform speed if $v = \frac{E}{B}$ (This is called as velocity selector condition)

Magnetic Force on a straight Current Carrying Wire

$$\vec{F} = I\,(\vec{L}\times\vec{B})$$

In general force is $\vec{F} = \int I(d\vec{l}\times\vec{B})$

Magnetic Interaction Force Between two Parallel Long Straight Currents

The force per unit length on either conductor is given by $F = \frac{\mu_0}{2\pi}\frac{I_1I_2}{r}$.

Where r = perpendicular distance between the parallel conductors

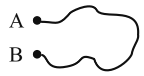
(*i*) Repulsive if the currents are anti-parallel.

(*ii*) Attractive if the currents are parallel.

Magnetic Torque on a Closed Current Circuit

When a plane closed current circuit of '*N*' turns and of area '*A*' per turn carrying a current *I* is placed in uniform magnetic field, it experience a zero net force, but experience a torque given by $\vec{\tau} = NI\vec{A}\times\vec{B} = \vec{M}\times\vec{B} = BINA\sin\theta$.

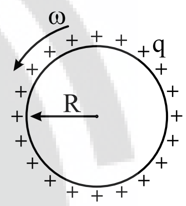
Force on a Random Shaped Conductor in a Uniform Magnetic Field



- Magnetic force on a closed loop in a uniform $\vec{B}$ is zero
- Force experienced by a wire of any shape is equivalent to force on a wire joining points A & B in a uniform magnetic field.

Magnetic Moment of a Rotating Charge

If a charge q is rotating at an angular velocity ω, its equivalent current is given as $I = \frac{q\omega}{2\pi}$ & its magnetic moment is $M = I\pi R^2 = \frac{1}{2}q\omega R^2$.



Galvanometer

Current sensitivity: $S_i = \frac{\alpha}{i} = \frac{NBA}{C}$

Voltage sensitivity (S_v): $S_v = \frac{\alpha}{V} = \frac{\alpha}{iR} = \frac{S_i}{R} = \frac{NBA}{RC}$